The documentation and process conversion measures necessary to comply with this amendment shall be completed by 2 July 1998.

INCH POUND

MIL-PRF-19500/596D
AMENDMENT 2
2 July 1998
SUPERSEDING
AMENDMENT 1
19 November 1997

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, REPETITIVE AVALANCHE, FIELD EFFECT,
TRANSISTOR, N-CHANNEL, SILICON,
TYPES 2N7218, 2N7219, 2N7221, 2N7222, 2N7218U, 2N7219U,
2N7221U, 2N7222U, JANTX, JANTXV, JANS, JANHC, AND JANKC

This amendment forms a part of MIL-PRF-19500/596D, dated 13 September 1996, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.3, maximum ratings, type 2N7221: Delete “300 V” and substitute “400 V”: Type 2N7222, delete “400 V” and substitute “500 V”.

PAGE 3

Figure 1, LL dimension, min. column : Delete “1.195” and substitute “.510”. Max column: Delete “1.235” and substitute “.570”.

PAGE 9

4.4.2 and 4.4.3, add the following: “and see 4.5.3, c = 0, n = 22 “

PAGE 10

* 4.5.2.b., after “3.3 A” add “minimum (2.5 A minimum for surface mount devices).”; 4.5.2.d., after “25 V” add “minimum (20 V minimum for surface mount devices).”

* The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Pages Replaced
5	5
6	reprinted without change

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PAGE 10

* 4.5.3, delete as added in amendment 1 and substitute as follows:

"4.5.3 Thermal impedance ($Z_{\theta JC}$ measurements). The $Z_{\theta JC}$ measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit (not to exceed .9°C/W or figure 4 thermal impedance curve) for $Z_{\theta JC}$ in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for engineering evaluation and disposition. This procedure may be used in lieu of an in line monitor. The following parameter measurements shall apply:

a. Measuring current (I_M): ..... 10 mA.

b. Drain heating current (I_H): ..... 3.3 A minimum (2.5 A minimum for surface mount devices).

c. Heating time (t_H): ..... 100 ms minimum (25 ms minimum for surface mount devices).

d. Drain-source heating voltage (V_H): ..... 25 V minimum (20 V minimum for surface mount devices).

e. Measurement time delay (t_{MD}): ..... 30 to 60 µs maximum.

f. Sample window time (t_{SW}): ..... 10 µs maximum."

PAGE 14

TABLE I, subgroup 7, gate to source charge test, limits max column, delete "12", "10.6", and "10.0" and substitute "16", "14.6", and "14.0" respectively.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961-2043-02)

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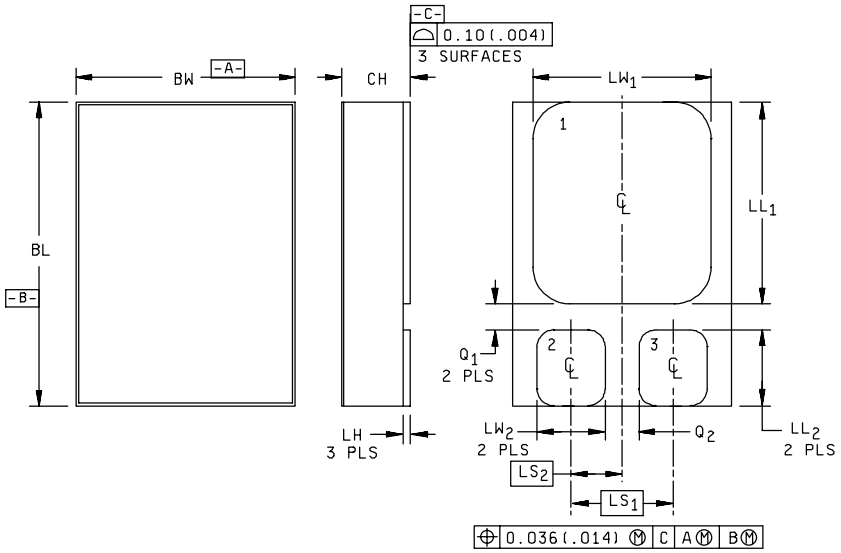

Dimensions				
Symbol	Inches		Millimeters	
	Min	Max	Min	Max
BL	.619	.631	15.73	16.02
BW	.444	.456	11.28	11.58
CH	- - -	.142	- - -	3.60
LH	.010	.020	0.26	0.50
LL_1	.409	.421	10.39	10.69
LL_2	.151	.163	3.84	4.14
LS_1	.210 BSC		5.33 BSC	
LS_2	.105 BSC		2.67 BSC	
LW_1	.369	.381	9.38	9.67
LW_2	.134	.146	3.41	3.70
Q_1	.030	- - -	0.76	- - -
Q_2	.035	- - -	0.89	- - -
Term 1	Drain			
Term 2	Gate			
Term 3	Source			

NOTES:
1. Controlling dimension: Inch.
2. Metric equivalents are given for information only.
3. The lid shall be electrically isolated from the drain, gate and source.
4. Dimensioning and Tolerancing shall be in accordance with ANSI Y14.5M 1982.

FIGURE 3. Dimensions and configuration of surface mount package outline .

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated performance specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 herein. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages.

3.3.1 Lead formation, material, and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-PRF-19500. Where a choice of finish is desired, it shall be specified in the acquisition document (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 and 100 percent dc testing in accordance with group A, subgroup 2 herein.

3.3.2 Internal construction. Multiple chip construction shall not be permitted.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500, except at the option of the manufacturer, the country of origin and/or the manufacturers identification may be omitted from the body of the transistor.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

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